Affine horizons and centered certificates for Diophantine consistency

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26 October 2026

Abstract

We give explicit polynomial equations whose lack of solutions in their specified domains is provably equivalent in ZFC to syntactic consistency of ZFC. The direct integer construction combines an affine proof horizon with a larger cofinal Pell frame and exact private-coordinate elimination. For natural variables, triangular changes of gate coordinates preserve nonnegative solutions while sharply reducing collected support and coefficient height. Centering a positive-domain carrier at a polynomial approximation to its extracted square root lowers its degree; a signed unsquared trace also improves two fourteen-integer circuits. The report supplies complete formulas, constructive witness maps, ordinary proofs and reproducible finite artifacts. It includes a separately calibrated pinned Start-marker4177 coefficient specialization and a general lemma for separating dependent radicals. No consistency decision, global optimum or new end-to-end kernel certificate is claimed.

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1 Objective and comparison boundary

For a fixed polynomial $P \in \mathbb{Z}[x_1, \ldots, x_n]$ and its specified domain D, our target is

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

The proofs establish solvability equivalent to a finite syntactic refutation, without asserting either consistency or a solution. The domains $\mathbb{Z}$, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0}$ are essential parts of the individual statements.

The immutable comparison snapshot of ProveIt is

`5274bc751a1566879ddc028a2d2a2af3c031654a`

It includes the new fixed-base modular, integral projection, reciprocal pooling, quotient-carrier and pooled-projection packets, as well as our preceding release. The baseline audit independently recollects the literal quartics and distinguishes placed work from incoming archives. In particular, the incoming direct quartic already reaches 216 integer variables. The 215-variable reciprocal companion has degree ten and must not be confused with a quartic.

A bounded late comparison stops at commit

`09764940c289c493aa573530eda86dd75f319787`

Its free-quotient circuit at 87 operations and degree 214 superseded our preliminary scaled-chart tradeoffs. The final ten-positive results below compose our centering with that stronger source. The late snapshot also contains 213- and 214-integer quartics with much larger coefficient heights; our 215-variable equations are therefore modest-coefficient tradeoffs, not the smallest inspected quartic dimension. Its maintained page uses Boolean-dummy4027 at 94,689 raw bytes. The checked 4177 tuple here is deliberately frozen, with its own calibration, rather than silently replaced by a later tuple.

Complexity measures are complete quantified variable count, exact total degree, quadratic residual count, collected support, maximum absolute coefficient, and live binary operation count. Addition, subtraction and multiplication each cost one, including squares and scalar multiplication; reuse is free. Circuit counts use integer leaves, not free rational constants. Fixed coefficients are not unknowns, but their complete coherent numerical tuple is part of every equation. These measures are not bit complexity and do not define a unique globally simplest equation.

Direct quartic over	Variables	Rows	Terms	Height
$\mathbb{Z}$, sparse	215	152	14,578	16,866
$\mathbb{Z}$, low height	215	152	34,700	4,512
$\mathbb{N}$, sparse	202	154	9,353	1,854,167
$\mathbb{N}$, low height	202	154	16,927	416,488
$\mathbb{N}$	203	155	6,524	1,979,888
$\mathbb{N}$	204	156	6,503	1,981,520

Table 1: Complete degree-four equations. Height is the largest absolute coefficient after collecting the entire sum of squared residuals.

2 Results and measured tradeoffs

The new sparse integer equation improves the incoming 216-variable, 153-row equation with 15,158 terms and height 334,952 in all four displayed measures. Against our preceding sparse 217-variable equation, it saves two variables, two rows and 37 terms at unchanged height. The low-height companion trades larger support for height 4,512. Larger-variable incoming equations with still lower heights remain genuine tradeoffs; no global coefficient minimum is claimed.

The sparse 202-natural-variable equation improves both the 9,818 terms and height 239,204,765 of its matched predecessor. The lower-height profile lowers the prior best displayed height 31,145,294 to 416,488 at greater support. These changes preserve the complete natural zero set through explicitly proved coordinate and residual maps.

A separate natural companion has 200 variables, 152 residuals, exact degree 8, 10,392 terms and height 13,654,244. It eliminates two auxiliary coordinates at a degree cost and is not a 200-variable quartic.

Domain	Variables	Degree	Operations	Matched comparison
$\mathbb{Z}_{>0}$	10	206	90	Late source 214/87
$\mathbb{Z}_{>0}$	10	210	89	Late source 214/87
$\mathbb{Z}$	14	85	110	Earlier 86/112
$\mathbb{Z}$	14	87	108	Earlier 88/110

Table 2: Integer-leaf circuit measurements. Comparison pairs are degree/operations at the same variable count.

The ten-positive-variable equations are new operation/degree tradeoffs: they lower degree but add operations relative to 214/87. The two fourteen-integer equations improve both degree and operation count over their matched parents. The incoming 14/82/113 point remains a different tradeoff, as do lower-variable equations of higher degree.

The complete Start-marker4177 specialization, credited to its incoming source, uses 98,219 raw bytes across four coherent coefficient ports; the largest has 196,790 bits. This is not a new encoding invented here. The new carrier front is separately bound to its complete native proof and finite controller certificate.

3 Affine pooled quartics in 215 unrestricted integers

Literal results. The two fully collected integer polynomials supplied with this section are

$$P_{215}(\boldsymbol{x}) = \sum_{i=0}^{151} R_i(\boldsymbol{x})^2, \qquad \boldsymbol{x} \in \mathbb{Z}^{215}.$$

The complete quadratic residuals, all collected quartic coefficients, and every coordinate map are included in the accompanying packet. The two residual bases give the following measured tradeoff:

Profile	Variables	Rows	Degree	Terms	Height
Sparse principal	215	152	4	14 578	16 866
Low-height alternative	215	152	4	34 700	4 512

Each residual coefficient has magnitude at most 64. All coordinates occur. The sparse form improves the prior 217-variable sparse equation in dimension, rows, and support at unchanged height. It also improves the incoming Pooled Projection 216-variable point, with 15 158 terms and height 334 952, in all four measures. The low-height form trades additional support for smaller coefficients. No global optimum is claimed.

Theorem and scope. For either literal polynomial, the ordinary proof below, composed with the retained numerical C222/ZF/ZFC refutation calibration, establishes in ZFC

$$\neg\exists \boldsymbol{x} \in \mathbb{Z}^{215}\ P_{215}(\boldsymbol{x}) = 0 \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This does not assert either side separately. The construction is direct through the numerical proof calculus, not K 215 simulation or Jones specialization. The named logical interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation
```

No new end-to-end proof-assistant compilation is claimed.

Attribution and new compatibility step. The immediate input is the retained Round 21 `beta217` equation. Source 160 already proves the affine-horizon, weak affine-frame, difference-error, and small-horizon estimates used here. Source 158 supplies fixed-base pooling and the shared-private-consumer mechanism; source 159 supplies the inherited radical projections. The new compatibility argument strengthens the Pell frame enough to make fixed-base pools work with absolute atom bounds instead of squared-atom bounds. The resulting composition saves two further coordinates and rows.

Weak atom bounds with a fourfold Pell frame

Set

$$t = 8h, \qquad d = t + 2, \qquad F = d^t, \qquad q = 4z + 1, \qquad y = F(2v + 1).$$

The powers here describe meanings of retained ordinary polynomial certificates, not primitive operations in the final equation. Change the old growth request from base t to base d, retaining its index t; replace the old square-horizon condition by the affine parameterization.

The 35 affine atoms consisting of the 34 retained actual atoms and the new base d use 43 pre-Pell coordinates u_j. They cover all bases, indices and outputs of all 17 requests. For integers a, u,

$$|au| \le u^2 + \lfloor a^2/4 \rfloor.$$

Consequently $L = c + \sum a_j u_j$ satisfies $|L| \le \sum_{a_j \neq 0} u_j^2 + |c| + \sum \lfloor a_j^2/4 \rfloor$. The exact maximum overhead is 21, attained by the affine derived-label expression; the new growth base has overhead 18. Thus the single row

$$W = 22 + \sum_{j=1}^{43} u_j^2 \tag{AF}$$

gives $W \ge 22$ and $|L| < W$ for every actual request atom. The full atom/port ledger is checked independently. No private Pell coordinate occurs in it. Importantly, this does *not* assert $L^2 < W$.

Replace the two old frame multiplication rows by

$$U = 4W(2z_f - 1), \qquad V = (4W+1)U,$$

and retain $A^2 + U^2 - V^2 = 1$. Pell classification at center $4W+1$ and $Y_\ell(4W+1) \equiv \ell \pmod{4W}$ force a nonzero index of magnitude at least $4W$. Hence

$$|A| > (4W)^{4W}, \qquad |A-1| > 4W^3. \tag{PG}$$

For completeness, the choice $\ell = 4W$ has $Y_{4W}(4W+1)/(4W)$ odd and supplies an integral z_f. The finite arithmetic front determines W before this private Pell choice, so there is no circular frame dependence.

Retain the entire signed semantic-anchor cap and every principal-pair norm. In the orientation pool replace only

$$(Y_{F+t} - F - t)^2 \quad \text{by} \quad (Y_{F+t} - Y_t - F)^2,$$

retaining $(Y_t - t)^2$ and all abscissa errors. The two error systems have the same zero set. As in source 160, first exclude the exceptional unit anchor ordinate by the cap and its nonzero orientation residue; only then apply the comparison-index theorem. Classified indices have magnitude below W. The reduced nonnegative orientation error satisfies

$$0 \le E_{\mathrm{or}} < 65W^2 + 60 < 4W^3 < |A-1|.$$

Its divisibility by $A-1$ forces every reduced original orientation error to vanish. Thus canonical signed Pell pairs are available before any arithmetic sign is assumed. This argument applies to either sign of A.

Pooling with absolute bounds

Let $r \le 15$, let $n \neq 0$, and suppose $|n|, |m_i|, |k_i| < W$, with canonical signed Pell pairs X_i, Y_i at A and retained gates $Z_i = AY_i$. Write

$$\begin{aligned} T &= 2An - n^2 - 1, \\ H &= \sum_i Y_i(m_i - X_i), \\ S &= \sum_i \bigl((m_i - X_i + Z_i)^2 - Y_i^2\bigr). \end{aligned}$$

Use only the two existing pooled coordinates K, L and quadratic rows

$$K - 2H - (2A - n)L = 0, \qquad S - nK + L = 0. \tag{PP}$$

The exact polynomial identity modulo the retained AY gates is

$$\sum_i (m_i - X_i + Z_i - nY_i)^2 = S - 2nH - T\sum_i Y_i^2.$$

Thus T divides the squared-error sum. Since $T \equiv -1 \pmod{n}$, negative indices can be cleared modulo T. Put $b = W - 1$. Then T divides

$$E = \sum_i (m_i n^b - n^{b+k_i})^2 \geq 0,$$

and every exponent displayed is nonnegative. The elementary bounds are

$$E \leq 60b^{4b} < (4W)^{4W}, \qquad |T| \geq 2|A| - (W-1)^2 - 1 > |A| > (4W)^{4W}.$$

Hence $E = 0$. At $|n| \geq 2$ this gives $k_i \geq 0$ and $m_i = n^{k_i}$; a negative index would imply $m_i n^{-k_i} = 1$. At $n = \pm 1$ it gives the exact signed unit value without asserting a nonnegative index. The $r = 1$ argument also revalidates all three singleton requests under the new weak frame. The converse uses the exact quotients $L = (S - 2nH)/T$ and $K = 2H + (2A - n)L$.

The two blocks have eleven requests at $q+1$ and three at q. This is a new size-bound composition of the credited pooling rows, not a new claim to those rows themselves.

Horizon, signs, and the inherited arithmetic budgets

For every integer h, the base $d = 8h + 2$ is nonzero and nonunit. Its canonical singleton certificate forces $t \geq 0$ and $F = d^t$. If $t = 0$, then $F = 1$ and y is odd. The endpoints read

$$x = 2 + (y+1)u_0, \qquad x = 1 + (y+1)u_t,$$

so $1 = (y+1)(u_t - u_0)$ would be even. Therefore $t \geq 8$ and $F \geq 10^8$.

The retained signed bootstrap is used in its noncircular order. Since $q = 4z+1$, first $|q+1| \geq 2$, so the $q+1$ pool gives nonnegative actual representatives and the nonzero semantic-anchor label. Hence $H \geq 2$; the row $B\mu = C + F + 1$ gives $B \neq 0$; the singleton at BH gives $x \geq 0$. Because y is a nonzero multiple of F, the last endpoint excludes $x = 0$. Thus $|q| \geq 2$, and only now the q pool gives $y + 1 \geq 0$, whence $y \geq F > 0$. The endpoints yield

$$x \geq F + 1, \qquad x \geq 2 + (t+1)y \geq 2 + (t+1)F.$$

Short signed-binomial recovery subsequently gives $B > 0$ and $q > 0$.

At the new smallest horizon one must not reuse $x > 2^{32}$. Instead $x \geq 900000002 > 2^{29}$; for $E_0 = 2^{23}$ and $2^s \leq x < 2^{s+1}$, $s \geq 29$,

$$E_0 \log_2(2x) < 2^{23}(s+2) < 2^s \leq x.$$

Thus the retained local-polynomial bound still lies below $2^x < N$. Also $F > 2^{22}$ supplies the derived-label budget. The selector inequality holds directly at $t = 8$ because $840^2 < 10^8$; at every other positive multiple of eight, $\Lambda_t^2 \leq 16^{t-1} < (t+2)^t = F$. The strict local ranges, all-primes CRT coverage and positive Horn-flow extraction therefore remain valid. Label multiplicities are

recovered at the still-larger evaluation base $F > t^t$. The inherited $y = F$ boundary is excluded after local recovery, not prematurely.

Conversely choose a cofinally large positive multiple of eight t beyond all finite proof parameters. Then $v_2(F) = t > v_2(t!)$; the inherited CRT construction supplies arbitrarily large admissible odd-square beta multipliers, which are allowed by the retained odd-linear condition. Choose even x and the parent's parity-prescribed representatives. The binomial B and H are odd, so $q = (BH)^x \equiv 1 \pmod 4$. After constructing all arithmetic wires, choose the weak frame and strong Pell frame, canonical principal witnesses, and the pooled quotients. This is a fresh completeness construction, not a map preserving the old square horizon or growth value.

An additional exact shared-consumer contraction

In the pinned 217-coordinate chart put

$$F = x_3,\ S = x_{12},\ S' = x_{13},\ a = x_8,\ b = x_9,\ s = x_{19},\ K = x_{62},$$

and $p = x_{75}, r = x_{76}, g = x_{77}, v = x_{78}, e_1 = x_{79}, e_2 = x_{80}$. The old rows include

$$r_K = K - FS', \quad r_p = p - 1 - Fb, \quad r_r = r - 1 - F(a-1), \quad r_g = g - 1 + 2F + 2Fa + 4Fb - 2Fs.$$

Delete p, r and their defining rows, and introduce only $J_{\mathrm{prod}} = FS$. Replace the three consumers by

$$\begin{aligned} C_{61} &= -S' - Kb + gv + e_1, \\ C_{64} &= -S - J_{\mathrm{prod}}(a-1) + v, \\ C_{65} &= S + J_{\mathrm{prod}} b - S' - K(2a + 4b + 2s) + e_2. \end{aligned}$$

Writing $r_J = J_{\mathrm{prod}} - FS$ and $D = 2a + 4b + 2s$, the exact identities are

$$\begin{aligned} C_{61} &= R_{61} + S' r_p - b r_K, \\ C_{64} &= R_{64} + S r_r - (a-1) r_J, \\ C_{65} &= R_{65} + b r_J + (8S' - S) r_p + 4S' r_r + S' r_g - D r_K. \end{aligned}$$

The forward map adjoins $J_{\mathrm{prod}} = FS$ and drops p, r; the inverse restores $p = 1 + Fb$ and $r = 1 + F(a-1)$. No other row uses the deleted coordinates. These maps are mutually inverse on integer zero sets. This is a specialized application of source 158's consumer method after earlier contractions, not a second credit for the already removed joint-square or third factor. Together with the affine horizon it changes 217/154 to 215/152.

Finite certification. The packet retains the complete parent archive, explicit source edits, all forward and backward maps, literal coefficients and frozen row operations. A separate checker uses only the standard library, imports no producer, and verifies complete ideal identities, all 17 actual request ports, the 43-coordinate majorant, every inverse row operation, all collected coefficients and coordinate occurrence. Corrupted frame, growth and restoration cases must fail. Both normal and optimized Python, read-only reconstruction and retained parent replay are supplied. These finite checks do not replace the unbounded proof or decide consistency.

4 Natural triangular syntax charts

The principal new quartic has 202 independently nonnegative integer variables and 154 quadratic residuals. Its selected support profile has 9 353 nonzero expanded monomials and coefficient height

1 854 167. A low-height profile on the same variables has 16 927 monomials and height 416 488. The construction is an exact natural-domain change of the retained round 21 system; it does not relax natural witnesses to signed integers or discard a Pell orientation condition.

Profile	Variables	Rows	Monomials	Height
Round21 principal	202	154	9 818	239 204 765
Round21 low height	202	154	12 706	31 145 294
New raw chart	202	154	9 652	13 654 244
New principal	202	154	9 353	1 854 167
New low height	202	154	16 927	416 488
New sparse companion	203	155	6 524	1 979 888
New sparse companion	204	156	6 503	1 981 520

The raw chart already improves both support and height over the preceding principal profile, and has lower height than its low-height companion. The last three refinements use independently certified integral residual shears. No optimality is asserted.

4.1 A natural-domain affine chart lemma

Let $R_1, \dots, R_r \in \mathbb{Z}[\boldsymbol{x}]$, and let S be a set of gate coordinates admitting an order $g_1, \dots, g_s$. Suppose the retained residuals include

$$x_{g_i} - f_i(\boldsymbol{x}) = 0, \qquad 1 \le i \le s, \tag{1}$$

where every coefficient of f_i is nonnegative, f_i does not contain x_{g_i}, and every coordinate from S occurring in f_i belongs to $g_1, \dots, g_{i-1}$. Coordinates outside S require no new interpretation.

Perform elementary changes, read sequentially,

$$x_v^{\text{new}} = x_v^{\text{current}} + L(\boldsymbol{x}^{\text{current}}), \tag{2}$$

where $v \in S$, the affine polynomial L has nonnegative integer coefficients and does not contain x_v. Let F be their composition and $G = F^{-1}$ its integral affine inverse. Then

$$F : \{\boldsymbol{x} \in \mathbb{N}^n : R_j(\boldsymbol{x}) = 0 \ \forall j\} \longrightarrow \{\boldsymbol{y} \in \mathbb{N}^n : R_j(G(\boldsymbol{y})) = 0 \ \forall j\} \tag{3}$$

is a bijection.

Indeed, every forward step preserves nonnegativity. Conversely, given a natural zero $\boldsymbol{y}$, put $\boldsymbol{x} = G(\boldsymbol{y}) \in \mathbb{Z}^n$. Coordinates outside S are unchanged and nonnegative. All original residuals vanish by substitution. Induction in (1) now proves $x_{g_i} = f_i(\boldsymbol{x}) \ge 0$. Both affine composite identities hold over $\mathbb{Z}^n$, so these maps are mutually inverse on the natural zero sets.

The inverse is emphatically not required to preserve the whole orthant: $x_v^{\text{current}} = x_v^{\text{new}} - L(\boldsymbol{x}^{\text{new}})$ can be negative at points that are not zeros. It is the retained triangular positive gate system that restores nonnegativity. Since the substitutions are affine, quadratic residuals remain quadratic.

4.2 The twelve explicit changes

Use the inherited formal labels, and abbreviate $a = \texttt{crt.encoded.a}$, $b = \texttt{crt.encoded.b}$, $G = \texttt{syntax.shared_shifted_square}$, $m_j = \texttt{mul}.j$, and $a_j = \texttt{add}.j$. The changes are

$$\begin{aligned}
m_{36} &\leftarrow m_{36} + 2b + 3,\\
m_{97} &\leftarrow m_{97} + 8b + m_{36} + 4m_{60} + 11,\\
m_{173} &\leftarrow m_{173} + 2a + m_{74} + 16G + 16,\\
m_{243} &\leftarrow m_{243} + 16m_{232} + 4m_{237} + 33,\\
m_{131} &\leftarrow m_{131} + 2a + 4m_{36} + m_{74},\\
m_{48} &\leftarrow m_{48} + 32m_{36} + 4G + 34,\\
m_{225} &\leftarrow m_{225} + 16m_{214} + 4m_{219} + 34,\\
m_{274} &\leftarrow m_{274} + 16m_{263} + 4m_{268} + 34,\\
m_{148} &\leftarrow m_{148} + 16m_{36} + 4G + 18,\\
m_{317} &\leftarrow m_{317} + 5m_{36},\\
m_{256} &\leftarrow m_{256} + 2a_{253} + 2,\\
a_{253} &\leftarrow a_{253} + a.
\end{aligned}$$

Every right-hand side means its current value at that step. In particular, the last change occurs after the change to m_{256}.

In the literal parent row numbering, starting at zero, the restoration order is

gate	m_{36}	m_{48}	m_{97}	m_{131}	m_{148}	m_{173}	m_{225}	m_{243}	a_{253}	m_{256}	m_{274}	m_{317}
row	6	8	14	18	20	23	30	33	35	36	39	45.

Each listed row is literally a monic output minus a polynomial with nonnegative coefficients, and translated inputs occur only at earlier positions of this order. The finite checker verifies these facts from the coefficient lists. No outer semantic coordinate, power denominator, Pell ordinate, CRT remainder or source value is translated.

For the 203-coordinate companion omit the absent targets m_{173} and m_{131}. For the 204-coordinate companion additionally omit m_{97}. All inputs in the remaining changes still exist; their own defining-row orders meet the same criterion.

4.3 A cancellation that explains the improvement

Let m and p denote the original m_{36} and m_{48}. The original m_{55} defining polynomial is

$$(2p + 128b + 64m + 8G + 259)(2p + 128b + 64m + 8G + 260). \tag{4}$$

The first change gives $\ell = m + 2b + 3$, and the sixth gives

$$z = p + 32\ell + 4G + 34 = p + 64b + 32m + 4G + 130.$$

Consequently (4) becomes simply

$$(2z - 1)(2z) = 4z^2 - 2z. \tag{5}$$

A dense quadratic row has become a one-variable quadratic without a new coordinate or a divisibility condition. Its negative linear coefficient is harmless: natural restoration uses the original

gate polynomial, not coefficientwise positivity of the transformed row. The other changes make analogous affine-accumulator cancellations.

For integral k and distinct i, j, an elementary residual change $R_i \leftarrow R_i + kR_j$ is reversed by subtraction. Thus every supplied residual-basis sequence preserves the common zero set. Over natural inputs the sum of squares is zero precisely when every residual is zero, so each newly squared basis has the same natural zeros. The numerical sum-of-squares polynomial itself need not remain equal. The principal, low-height, 203- and 204-variable profiles use respectively $24, 53, 25, 26$ such reversible row changes.

4.4 A separate degree-eight dimension tradeoff

The raw chart also yields a 200-variable polynomial of exact degree eight with 152 residuals, 10 392 expanded monomials and height 13 654 244. This is not a quartic-dimension improvement.

Write $n = q > 0$, $A = \texttt{shared.lower_A} + \texttt{shared.A_slack}$, and $T = 2An - n^2 - 1$. Retain the three inherited pooled pairs (Y_i, Z_i) and affine output expressions m_i, and retain the natural witness M. The three pooled quadratic rows are

$$\begin{aligned} R_0 &= U - \sum_i Y_i^2, \\ R_1 &= K - 2\sum_i Y_i(m_i + Z_i) - (2A - n)M + nU, \\ R_2 &= \sum_i (m_i + Z_i)^2 + M - nK. \end{aligned}$$

Neither U nor K occurs elsewhere. Delete them and these three rows, and retain the single quartic residual

$$Q = \sum_i (m_i + Z_i - nY_i)^2 - TM = R_2 + nR_1 - n^2 R_0. \tag{6}$$

Projection takes every natural zero of the quadratic system to a natural zero of the new system. Conversely, first restore the syntax gates and use the unchanged unpooled power authentication to obtain $n > 0$, before interpreting any pooled power. Set

$$U = \sum_i Y_i^2, \qquad K = 2\sum_i Y_i(m_i + Z_i) + (2A - n)M - nU.$$

The first expression is natural and the second is integral. Equation (6) gives $nK = \sum_i (m_i + Z_i)^2 + M \geq 0$; hence $K \geq 0$. Thus both eliminated witnesses are restored in the natural domain. Integrality of K comes from its polynomial formula, not an unproved divisibility assertion.

4.5 Calibration and exact verification

The gate restoration recovers the entire inherited residual system before any Pell or semantic theorem is used. Its strengthened lower frame, additive frame budgets, and inverse-power argument for the $y + 1$ seed are unchanged. In particular no deleted orientation is silently re-assumed. The inherited finite-refutation theorem therefore transfers to every profile above. With the fixed effective ZFC presentation and its retained C222/ZF/ZFC calibration, ZFC proves that each displayed no-natural-zero sentence is equivalent to Con(ZFC).

The package contains every residual and every collected integer coefficient. An independent standard-library checker uses sparse exponent maps to verify both affine compositions, every transformed row, the original positive gate definitions and their dependency order, all sum-of-squares

coefficients, and the elimination identities. Separate checks replay and reverse every residual-basis change and reject corrupted certificates. A clean rebuild reproduces the coefficient files, and the printed TeX residuals are parsed back to their exact polynomials. Normal and optimized Python executions both pass. These finite computations check exact algebra; they are not represented as an end-to-end proof-assistant formalization of the unbounded inherited theorem.

5 Centered free quotient carriers in ten positive integers

The complete literal polynomials here use ten independently strictly positive integer unknowns, with operation/degree pairs

$$89/210 \qquad \text{and} \qquad 90/206.$$

All binary additions, subtractions, scalar multiplications, and squares are charged; leaves are integers. The bounded late source 172 at ProveIt `09764940c` supplies 87/214. The new pairs exchange two or three gates for four or eight degrees. They are not simultaneous improvements or global minimum claims.

The earlier scaled-chart 87/228 and 88/224 artifacts remain supplied as verified method controls, but source 172 strictly dominates them in operation/degree at the same historical tuple. Their smaller-coefficient 4177 specializations are a separate storage comparison. They are no longer headline operation/degree frontier points. Source 172's free quotient $Y = s$, scale $\lambda = q$, and index-three-compatible transport proof are credited to that source and its retained predecessors. It does not already perform the polynomial centering used here.

The whole historical radix 67 tuple remains little-endian. Each new front also has a separately calibrated whole Start-marker 4177 specialization: 98,219 raw bytes, largest port 196,790 bits, minimal unsigned big-endian. This is the pinned tuple at the earlier `5274bc751` snapshot, not a global storage record: the bounded late source 174 supplies a distinct 4127 encoding at 97,046 bytes; the maintained Boolean-dummy 4027 page uses 94,689 bytes. No ports or controller certificates are mixed.

The result is an ordinary proof composition at the explicit retained native/refutation boundary. It is not a consistency decision, a claimed zero, a new formal-kernel object, or an encoded internal-ZFC replay.

5.1 The complete free-chart equations

The ten independent inputs are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

For the selected entire fixed tuple β, K_0, M_C, M_F, define

$$h = \beta J, \quad q = h + 1, \quad C = q - F - \alpha,$$
$$X = qw, \quad Y = s, \quad \lambda = q, \quad k = \eta + \zeta, \quad c = kY + \eta,$$
$$A = qY(X + 1) + 1, \quad \Delta = A^2 - 1, \quad H = 4A - 5,$$
$$v = Xq^2Y^2, \quad \ell = vk, \quad E = 1 + \ell(\ell + k),$$
$$R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J,$$
$$B = (K_0 + w)C - F, \quad U = ic^2, \quad a = \Delta U^2 E,$$
$$F_0 = 1 + a, \quad G = EF_0, \quad f = a + F_0, \quad Q = 4\Delta a F_0,$$
$$b = cT - R, \quad V = fb - c, \quad K = X(3c - 2X) + 2,$$
$$e_6 = 6(k - R - 1) - \ell(k^2 - 4), \quad D = 6q^7, \quad M = hDH.$$

Choose one polynomial center,

$$\widehat{g} = AUE \quad (90/206), \qquad \widehat{g} = (A + 1)UE \quad (89/210).$$

Then set

$$Z = hD\{H[n - V(bB + \widehat{g})] + K\} - H[DB + h(e_6 - 6q^4 s)], \tag{7}$$
$$P = Q(Z^2 - M^2GV^2) + (M^2G - Z^2). \tag{8}$$

The equation is $P = 0$. All named quantities after the ten inputs are polynomial abbreviations, not further witnesses. There are no rational leaves, divisions, recurrences, binomials, or variable-exponent gates.

For proof only let $L = bV$, $W = 1 + hL$, $t = B/h$, $\kappa = K/H$, $e^* = e_6/6$, $\rho = e^*/q^7$, $\sigma = Y/q^3$, and $z = Z/M$. Before asserting integrality one has the exact rational identity

$$z = n + \kappa + \sigma - Wt - \rho - \widehat{g}V. \tag{9}$$

If Z_0 is the uncentered source 172 numerator, then $Z = Z_0 - M\widehat{g}V$ identically.

5.2 Soundness, including the auxiliary index-three branch

The complete source 172 proof is retained in `late172/parent/`. Its root extraction and signed all-index folding use the final norm, not the specific carrier numerator. At a zero they give positive integral τ, f_0, g, y and $\epsilon = \pm 1$ with

$$E = \tau^2, \quad F_0 = f_0^2, \quad g = \tau f_0, \quad z = \epsilon g y, \quad QV^2 - (Q - 1)y^2 = 1,$$
$$f^2 - \Delta I^2 = 1, \quad I = 2Ug, \quad c^2 \mid I, \quad c = \psi_A(p), \quad p \geq 1 \text{ odd}.$$

Since $c \geq 3$, $p \geq 3$ and $c \geq c_3 = 4\Delta + 3 > 4q^4$, hence $K > 0$. This works even at $Y = 1$: no assumption $q \mid Y$, exclusion of $p = 3$ using that assumption, or exclusion of auxiliary index 3 is imported.

The source's trace-linear argument gives nonzero b, V of the same sign, and

$$L = |b||V| > 0, \qquad y < \tfrac{3}{2}|V|, \qquad f > 6g, \qquad |b| > 4\Delta f - 1.$$

Although the source displayed the weaker consequence $|b| > 4g$, its actual preceding bound gives, using $\Delta \geq 3$ and integral $g \geq 1$,

$$|b| > 24\Delta g - 1 \geq 72g - 1 > 8g. \tag{10}$$

For $g_0 = AUE$, the identity $g_0^2 - G = U^2E^2 - E$ gives $0 < g_0 < 2g$. For $g_+ = (A+1)UE$,

$$g_+^2/G < (A+1)^2/\Delta = (A+1)/(A-1) < 2,$$

so also $0 < g_+ < 2g$. Therefore either center satisfies

$$gy + \widehat{g}|V| < \tfrac{7}{2}g|V| < \tfrac{7}{16}L < L/2. \tag{11}$$

Since $k \geq 2$, $e^* \leq k - R - 1 < b$, hence $\rho < |b|/q^7$. If $B \leq -1$, then $Wt < -L$. Positive κ, σ and (9) give

$$n < L/2 - L + |b|/q^7 < 0,$$

a contradiction. Thus $B \geq 0, C > 0$, before assuming any sign of R. The whole-tuple mask bound now gives $3q < R < q^4 - 2$. As $c > 2R$, $b, V > 0$. The unchanged normalized-rank and signed auxiliary-index theorem yields $p = R$, followed by R odd, q even, and J odd.

5.3 Authentication, projection, and positive completeness

Because $Z - Z_0$ is a multiple of $M = hDH$, every numerator reduction modulo D and H in source 172 remains unchanged. First $k = 2\psi_{1+2v}(m)$ with $m \geq 2$, and literal cancellation of 6 gives $q^4 \mid e^*$. The full recurrence and cubic-difference identity

$$3[e^* - (2m - R - 1)] = 4v^2(24Z_m - A_m S_m)$$

with $v = q^3 ws^2$ give $q^4 \mid [e^* - (2m - R - 1)]$ at every prime, including 3. Monotonicity gives $m < R$, hence $m = (R+1)/2$. No q^7 divisibility is inferred yet.

Reduction modulo H gives $H \mid 6hK$. The ratio theorem is applied in its actual uniform family $\lambda = q, Y \geq 1$, giving $Y > X^r/3$ for $R = 2r + 1, r \geq 48$. The universal projector and whole-product height bound then yield $X = 2^R$ without canceling an unproved factor of 6 h. Now q is a power of 2; rank growth gives $q^6 \mid X$, hence $q^8 \mid v$. Only at this point does the cubic identity give $q^7 \mid e^*$. Thus ρ, κ are integral. Since $\widehat{g}V$ is integral on every input, the remaining carrier yields $q^3 \mid Y$ and $h \mid B$, so $t \geq 0$ is integral. The retained fixed-minus parity, uniform half-binomial rounding, and fresh native reconstruction at $Y(X+1)+2$ then give the actual native words and a finite syntactic ZFC refutation.

Conversely the separately calibrated parent construction supplies a positive zero with the free chart $Y = s, \lambda = q$ and $V > 0$. Keep all inputs except n and define

$$n_{\text{new}} = n_{\text{old}} + \widehat{g}V > 0.$$

The center is independent of n, and exactly

$$n_{\text{new}} - V(bB + \widehat{g}) = n_{\text{old}} - bVB.$$

Thus the numerator and final polynomial value are preserved. No new cofinal seed or Pell-branch choice is needed. This is a positive map on constructed parent zeros; soundness is independent because an arbitrary inverse shift need not stay positive.

5.4 Literal costs and all highest-degree terms

The source uses 87 gates. Replacing its three gates for $L = bV$, LB, $n - LB$ by four for bB, $bB + \widehat{g}$, $V(bB + \widehat{g})$ and $n - V(bB + \widehat{g})$ adds one. For AUE, computing $A = (A - 1) + 1$ and $A(UE)$ adds two more, giving 90. For $(A + 1)UE$, $A + 1$ is already live, so only one more gate is needed, giving 89.

The exact cancellation is

$$Z^2 - M^2GV^2 = Z_0^2 - 2M\widehat{g}VZ_0 + M^2(\widehat{g}^2 - G)V^2,$$

where

$$(AUE)^2 - G = U^2E^2 - E, \qquad ((A + 1)UE)^2 - G = 2(A + 1)U^2E^2 - E.$$

The actual free-chart degrees are

$$(\deg Q, \deg M, \deg G, \deg V, \deg Z_0, \deg U, \deg E, \deg \widehat{g}) = (72, 12, 46, 36, 54, 5, 14, 23).$$

For the first center, the three inside terms have degrees 108,125,134, so multiplication by Q gives 206. For the second, the last is 138, giving 210. The final correction has degree 142 in both cases.

Put $q_0 = \beta J$, $k_0 = \eta + \zeta$. The complete leading forms are

$$[P_A]_{206} = 9 \cdot 2^{10} q_0^{82} w^{24} s^{56} i^{10} k_0^{32} (q_0 - F)^2,$$
$$[P_{A+1}]_{210} = 9 \cdot 2^{11} q_0^{84} w^{25} s^{57} i^{10} k_0^{32} (q_0 - F)^2.$$

Each has 99 terms and is nonzero for actual $\beta > 0$. Here the leading term of b is $-q_0^3(q_0 - F)$; the scaled chart's T-dependent tie must not be copied. Independent exact cuts cover every actual gate, prove the cancellation, and check all 99 terms. All 180 or 178 alternative gate-operation corruptions are rejected for each complete tuple.

5.5 Whole-tuple calibration, recovery, and checks

The source 172 closure was reconstructed at the bounded late pin using 59 exact PLACEMENT aliases; all 67 inherited dependency pins match. Its builder and full 508-mutation deep checker pass. Normal/optimized checks for the four new graphs also agree. The historical tuple and its full finite recipe remain unchanged; the entire 4177 tuple, actual physical support and Pcs are separately reconstructed.

For 4177, there are 980 physical positions but 977 nonzero weights. Marker 1048 has zero weight and remains physical; the separate kernel monomial at 1048 copies Start selector 0. Every one of the 3297 coefficient slots and all four integers are checked, with $M_F = M_{F,\text{native}} + \beta$ exactly once. The 4177-specific Pcs uses primes 4177/224113, orders 23/87, spacing 2001, and nonzero H residues 9/2620/158694 modulo 47/4177/224113.

The free-chart endpoint is exactly $X = 2^R, q^3 \mid Y, C > 0$ and $(K_0 + w)C - F = (q - 1)t$ with integral $t \geq 0$. It reconstructs fresh source witnesses at $Y(X + 1) + 2$, decoded by the 4177 native proof. Conversely that proof provides fresh canonical outer data to which the source 172 $\lambda = q$ construction and our positive centering map apply. Neither old words nor old controller certificates are substituted.

The frozen 4177 gzip blobs are unchanged. The inherited page builder has the explicitly recorded runtime OS-header 03/ff difference; raw integer reconstruction, stored hashes, new graph bytechecks and Pcs pass. We do not claim that the runtime-sensitive inherited compressed bytecheck passed.

At each whole retained calibration, ZFC proves absence of a strictly positive zero equivalent to Con(ZFC). Finite checks are separate from the unbounded Pell and native theorems. This is an ordinary composed result, not a new end-to-end formalization or a consistency proof. Full proofs, original attribution, licenses and finite recipes accompany every numerical specialization.

6 Even strong traces and unweighted first roots

Two new fixed ordinary polynomials have fourteen unrestricted integer unknowns:

$$(\deg, \mathrm{ops}) = (85, 110), \qquad (87, 108).$$

The charged binary-operation counts are respectively $(63, 24, 23)$ and $(62, 23, 23)$ for multiplication, addition and subtraction. They strictly improve the matched $(86, 112)$ and $(88, 110)$ points; $(82, 113)$ remains a separate tradeoff. All six full fixed numeral payloads and their finite reconstruction recipe are retained. These are ordinary reductions through the inherited native/compiler/refutation theorem, not a consistency assertion, global optimum, or new end-to-end proof-assistant certificate.

6.1 Complete polynomial definitions

The unknowns are

$$j, C, z, h, r, v, w, T, c, n, f, y, p_2, p_3 \in \mathbb{Z}.$$

Use the authentic ports $\beta, K_0, M_C, M_F, A_c, B_c$, with M_F already shifted, $\beta = 2^{1064921}-1$, $A_c = 2^{1113}$ and $B_c = 2^{666104}$. Define

$$\begin{gathered}
J = 6j+1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \quad X = qW, \quad Y = v, \\
F = (K_0 + W)C - zq_m, \quad R = [q(q-F) - C](q^2-1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \quad \ell = XY^2, \\
\eta = c - Yk, \quad L = \ell k, \quad I = (8r+4)c^2, \quad Q = \Delta(f^2-1), \\
V = f(cT - R) - c, \quad M = 2q^3, \quad U = MH, \quad K = X(2X - 3c) - 2, \\
Z_0 = Un + MK + H(Y - q^3), \quad D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2.
\end{gathered}$$

Choose the chart

deg	k	$\mathcal{B}$	Z
85	$R + 1 + h\ell$	$(Yk)^2(q^2 - D_{\mathrm{err}}) - q^2[4(Y\eta)^2 + (kq^2)^2]$	$Z_0 + UL$
87	$R + 1 + hE$	$(Yk)^2(q^2 - D_{\mathrm{err}}) - q^2[4(Y\eta)^2 + (kq^3)^2]$	Z_0

and finish with

$$\begin{aligned}
G &= \mathcal{B} - 1 - p_2^2 - p_3(p_3+1), & S &= f^2 - \Delta I^2, \\
u_u &= (V+y)^2, & E_a &= u_u - Q(u_u - V^2), \\
D_\tau &= 1 + L(L+k), & & \\
A &= U^2 G, & B &= U^2 D_\tau, \\
N &= (Z^2 - A - B)^2 - 4AB, & P &= (N+1)SE_a - 1.
\end{aligned}$$

The identity $E_a = V^2 - (Q-1)y(2V+y)$ shows that sharing u_u saves one charged multiplication without changing the polynomial. Every name denotes a polynomial abbreviation, not an additional unknown. The 87-chart evaluator uses $L = E(Yk)$ and does not separately compute ℓ.

6.2 Unweighted extraction, including all exceptional branches

Unconditionally q is even and nonzero, $|W| \geq 4$, $|X| \geq 8$, H is odd and nonzero, and $U \neq 0$. Since $a = (X+1)Y$ is zero or has $|a| \geq 7$, $\Delta = (a+2)^2 - 1 \geq 3$. Also ℓ is even, so $D_\tau = 1 + \ell(\ell+1)k^2 \geq 1$.

At $P = 0$ the three integer factors are units. As N is a square modulo four, $N + 1 \neq -1$; as $4 \mid I$, $S \equiv f^2 \pmod{16}$ and $S \neq -1$. Therefore $N = 0$, $S = 1$, $E_a = 1$. With $\rho = Z/U$,

$$(\rho^2 - G - D_\tau)^2 = 4GD_\tau.$$

This forces $G \geq 0$, including $G = 0$. Hence $\mathcal{B} > 0$ and $Yk \neq 0$. Since $|\ell| \geq 8$ and $|X| \geq 4|q|$,

$$D_\tau \geq 1 + \tfrac{1}{2}X^2Y^4k^2 \geq 1 + 8q^2Y^4k^2 > q^2Y^2k^2 \geq \mathcal{B} > G.$$

Thus $\rho \neq 0$ before any division by it. The rationals

$$p = \frac{\rho^2 + G - D_\tau}{2\rho}, \qquad \tau = \rho - p$$

satisfy $p^2 = G$, $\tau^2 = D_\tau$. Each is integral because a rational with integral square is integral. This proof covers zero and coincident square classes; no generic independence hypothesis occurs. The universal signed factorization is

$$(x^2 - A - B)^2 - 4AB = \prod_{\epsilon_1, \epsilon_2 \in \{\pm 1\}} (x - \epsilon_1\sqrt{A} - \epsilon_2\sqrt{B}).$$

Write $s = 1$ for degree 85 and $s = 0$ for degree 87. Restore the even-trace fifteen-variable kernel by

$$p_1 = p, \qquad n_{\text{old}} = n + sL - (2L + k) - p + \tau.$$

Its carrier is

$$\mathcal{A} = (Z_{\text{old}} + Uk)^2 + 4U(LZ_{\text{old}} - U) = (Z_{\text{old}} + U(2L + k))^2 - 4U^2D_\tau,$$

and therefore vanishes. The remaining rows are $S - 1 = 0$, $E_a - 1 = 0$ and $\mathcal{B} - 1 - p_1^2 - p_2^2 - p_3(p_3+1) = 0$. Conversely any zero of this kernel has an integral τ with $Z_{\text{old}}/U + 2L + k = 2\tau$: the rational on the left has square $4D_\tau$, so is an even integer. Delete p_1 and set

$$n = n_{\text{old}} - sL + (2L + k) + p_1 - \tau.$$

Then $\rho = p_1 + \tau$ and $P = 0$. These are total maps on zeros, not claimed polynomial automorphisms; the root-dependent integral shift removes one gate.

6.3 Signed unsquared trace: the new parity argument

The comparison signed kernel has odd strong scale $(2r_{\text{old}} + 1)c^2$ and squared trace $f_{\text{old}}^2(cT_{\text{old}} - R) - c$. Its retained proof is `inherited_round20/inherited/PROOF.md` for the q^6/XY^2 chart, and `FREE_SCALE15_PROOF.md` in its inherited roots source for q^8/XY. The normalized rank and auxiliary lemmas are in the same source's `06_PROOF.md`. We verify the changed hypotheses before restoring a literal old zero.

The unchanged carrier and guard give $c \neq 0$, the strict centered strip, $q_0 = |q| \geq 16$, $x = |X| \geq 4q_0$ and

$$q_0(q_0^2 - 1) < R < q_0^4 - 2.$$

They give $|Y| > q_0^2$, $|XY^2| > 3R+1$ in the 85 chart, and $|Y| > q_0^3$, $|XY| > 3R+1$ in the 87 chart. Carrier integrality gives even Y and the exact divisibilities $M \mid H(Y-q^3)$, $H \mid MK$ without premature cancellation. Put $B_0 = |a+2|$, $c_0 = |c|$, $I_+ = |8r+4|c_0^2$ and $f_+ = |f|$. Then $I_+ > 0$ is even, $B_0 \geq 3$, and

$$f_+^2 - (B_0^2-1)I_+^2 = 1, \qquad f_+ > 2I_+.$$

Pell classification gives $f_+ = \chi_{B_0}(m)$, $I_+ = \psi_{B_0}(m)$. The residue $V \equiv -c \pmod{f_+}$ is unchanged. The ordinary odd auxiliary index and folding lemma therefore recover $c_0 = \psi_{B_0}(p)$ before any residue modulo c_0 is needed: fold the auxiliary index to $0 \leq p \leq m$, and use $\psi_{B_0}(p)+c_0 \leq 2I_+ < f_+$ to turn its signed congruence into equality. The first Pell classification and centered bounds give $p > n_0 \geq 24$ and $c_0 > 2p, 2R$.

The normalized rank lemma gives $m = pv$ and $c_0 \mid v$. Indeed, strong divisibility gives $p \mid m$; with $D = \chi_{B_0}(p)$, composition gives $I_+ = c_0\psi_D(v)$, and $\psi_D(v) \equiv vD^{v-1} \pmod{c_0}$ with $\gcd(D, c_0) = 1$ gives $c_0 \mid v$. Now $\psi_{B_0}(m) \equiv m \pmod 2$, so m is even. If c_0 is even, v is even; if c_0 is odd, p is odd and $m = pv$ again makes v even. As $D^2 \equiv 1 \pmod{c_0}$ and $\chi_D(v) \equiv D^v \pmod{c_0}$,

$$|f| = \chi_D(v) \equiv 1 \pmod{c_0}.$$

Consequently the unsquared trace has exactly the required signed residues

$$V \equiv -c \pmod{|f|}, \qquad V \equiv \pm R \pmod{c_0}.$$

The signed auxiliary step-down gives an odd index ℓ with $\ell \equiv \pm p \pmod m$ and $\ell \equiv \pm R \pmod{c_0}$. Since $c_0 \mid m$ and $0 < p, R < c_0/2$, it follows that $p = R$. Thus the extra sign does not invalidate the argument. The unchanged surviving first-index branch gives $2n_0 = R+1$. Whole-product projection, the modulo-three orientation and late integer cancellation then give

$$c, k, Y > 0, \quad X = 2^R, \quad R \text{ odd}, \quad Y = (2v_{\text{scale}}+1)q^3, \quad H \mid K, \quad 0 < \eta/k < 1/4.$$

These steps use only the already established indices, carrier and strip, not the old square on f. In particular $B_0 = a+2 \equiv 2 \pmod 8$ and c is odd.

6.4 Fresh strong witnesses: soundness and completeness

One cannot keep f, T when returning to an odd-scale source. Let

$$m_0 = Rc, \qquad f_0 = \chi_{B_0}(m_0), \qquad I_0 = \psi_{B_0}(m_0).$$

Composition gives $c^2 \mid I_0$; as R, c, m_0 are odd, I_0/c^2 is odd. Thus $r_{\text{old}} = (I_0/c^2 - 1)/2$ restores the old strong row. For every odd $R = 2v+1$, the integer quotient polynomials obey

$$\chi_Z(2v+1) = ZQ_v(Z^2), \quad Q_v(0) = (-1)^v(2v+1), \quad Q_v(1-B_0^2) = (-1)^v\psi_{B_0}(2v+1).$$

They follow from $Q_0 = 1$, $Q_1 = 4x-3$ and $Q_{v+1} = (4x-2)Q_v - Q_{v-1}$. Set $S_0 = \Delta I_0$ and

$$V_0 = (-1)^{v+1}Q_v(S_0^2), \qquad u_0 = \psi_{S_0}(R).$$

Then $S_0^2V_0^2 - (S_0^2-1)u_0^2 = 1$, while $V_0 \equiv -c \pmod{f_0^2}$ and $V_0 \equiv -R \pmod c$. Since $\gcd(c, f_0) = 1$ and $f_0^2 \equiv 1 \pmod c$,

$$T_{\text{old}} = \frac{V_0 + c + Rf_0^2}{cf_0^2}, \qquad y_{\text{old}} = u_0 - V_0$$

are integers and restore the squared auxiliary trace. Every outer, carrier and guard coordinate is unchanged. This is a literal zero of the correct old signed kernel (separately for each chart), not an assumed $R \equiv 3 \pmod 4$ branch. Its authenticated native/compiler theorem applies.

Conversely every old zero has the same established outer data. Double the seed:

$$f = 2f_0^2 - 1, \qquad I = 2f_0 I_0.$$

Because $B_0 \equiv 2 \pmod 4$ and m_0 is odd, $f_0 \equiv 2 \pmod 4$. Hence $I/(4c^2) = (f_0/2)(I_0/c^2)$ is odd, so $r = (I/(4c^2) - 1)/2$ is integral and $I = (8r+4)c^2$. The doubled pair has norm one; $f_0^2 \equiv 1 \pmod c$ directly gives $f \equiv 1 \pmod c$. Using the same signed quotient construction at $S = \Delta I$ yields V_*, u_*; now

$$T = \frac{V_* + c + Rf}{cf}, \qquad y = u_* - V_*$$

are integers because the numerator is divisible by the coprime integers c, f. This gives $V = f(cT - R) - c = V_*$ and the new auxiliary row. No positivity of T is required on the unrestricted-integer domain. Apply the unweighted root map to finish. Both directions are defined on all zeros using fresh witnesses.

6.5 Attained degrees and exact coefficients

Put

$$q_0 = 6\beta j, \quad X_0 = 12q_0 w, \quad Y_0 = v, \quad a_0 = X_0 Y_0,$$
$$F_0 = 12wC - q_0 z, \quad R_0 = -q_0^3 F_0, \quad \ell_0 = X_0 Y_0^2, \quad U_0 = 8q_0^3 a_0.$$

Both complete strong/auxiliary tops are

$$[S]_{12} = -64 a_0^2 r^2 c^4, \qquad [E_a]_{15} = 2a_0^2 f^3 R_0 y.$$

For degree 87,

$$[N]_{60} = U_0^4 R_0^4 (q_0^8 + \ell_0^2)^2, \qquad [j^{56} w^8 v^8 z^5 r^2 c^4 f^3 y] P_{87} = -2^{91} 3^{64} \beta^{56} \neq 0.$$

For degree 85 retain $k_0 = R_0 + h\ell_0$ and define

$$T_0 = A_c^2 Y_0^2 F_0^2 + 4q_0^2 Y_0^4 + q_0^6.$$

The entire cancellation

$$(Z_0 + UL)^2 - U^2 D_\tau = Z_0^2 + 2ULZ_0 - U^2(1 + Lk)$$

has degree 26. Thus the square in N has degree 56 and the other term has top

$$[N]_{58} = 4U_0^4 \ell_0^2 k_0^4 T_0, \qquad [j^{48} w^{10} v^{12} z^5 r^2 c^4 f^3 y] P_{85} = -2^{89} 3^{58} \beta^{48} \neq 0.$$

These full leading forms, with respectively 18 and 70 monomials after expansion, prove the exact degrees on the supplied fixed numeral slice, not merely generic or sampled degrees.

6.6 A separate correlated-root separation lemma

Let $F, D \equiv 1 \pmod 3$, $G \in \mathbb{Z}$, $3 \mid Q$ and $b, c \in \mathbb{Z}$. For algebraic roots

$$f^2 = F, \quad \tau^2 = D, \quad t^2 = \frac{Q(fb-c)^2 - 1}{Q-1}, \quad g^2 = G,$$

a rational $\rho = f + 3\tau + 9t + 27g$ forces all four roots rational. Work in the integral closure of $\mathbb{Z}_{(3)}$ in a normal field containing all roots. They are integral over this ring, since $Q - 1$ is a unit and $t^2 = \alpha + \beta f$ with $\alpha, \beta \in \mathbb{Z}_{(3)}$. If an automorphism flips f, subtract the trace equations to obtain $f = 3u$ with integral u. But $u^2 = F/9$ is rational and integral over $\mathbb{Z}_{(3)}$, hence lies in that ring, contradicting $F \equiv 1 \pmod 3$. Thus all automorphisms fix f, so f is rational. Divide the conjugate difference by three to repeat for τ; then t^2 is rational and a three-adic unit, so divide by nine and repeat for t. Finally g is rational. This includes imaginary roots and dependent square classes. A valid norm must use the different auxiliary roots associated to each sign of f; the independent four-root norm cannot simply be reused. The explored twelve-variable evaluators remained dominated, so no new twelve-variable frontier claim is made.

6.7 Verification and dependency boundary

The standard-library checker binds every literal register independently, all inputs and six ports, topology, coverage, liveness, exact operation counts, four-factor identity, root maps, doubling, full leading cancellations and attained coefficients. It rejects individual mutations of all 110 and 108 live gates. Normal and optimized Python runs pass the same explicit conditions. The finite reconstruction recomputes all 615883 coefficient bytes, 943 tiles and 69230 basis convolution checks. All 212 inherited source files are pinned. Recurrence samples are diagnostics, not proofs of the unbounded lemmas above. The current comparison snapshot is `5274bc751`; original source attributions and licenses are preserved. The ordinary native/compiler/refutation theorem remains the named inherited dependency, not a claim inferred from finite checks.

7 Reproduction and mathematical scope

The direct equations are sums of the squares of literal residual lists. Their complete collected coefficients and readable expressions are supplied. The compact equations are fully specified polynomial circuits with all fixed numeral payloads and reconstruction recipes. Proof-only factorials, Pell sequences, fractions, inequalities and variable exponents are not hidden primitives of any final polynomial.

The unbounded proofs establish sign recovery, divisibility, cofinal completeness, source restoration and domain preservation. Independent finite checkers bind them to actual rows and gates, full leading forms, coefficient bytes and inverse identities. Finite test cases supplement those proofs; they do not replace them. The portable replay runs in independent temporary copies under both normal and optimized Python and verifies that all sealed bytes remain unchanged. The final extracted ZIP is replayed separately.

Logical equivalence retains the specified numerical-calculus or native compiler and syntactic-refutation calibration. These inherited interfaces are explicit, not inferred merely from numerical coefficient bounds. No transitive proof-assistant build, encoded object-language ZFC derivation, consistency decision or global lower bound is claimed. The independent reviews are separate checks in this AI-assisted workflow, not outside human endorsements. No upstream repository is modified.

The Start-marker4177 tuple has its own complete native proof, physical support and finite controller certificate. Zero coefficient weight does not mean a physical marker may be removed. Its four ports are not mixed with earlier specializations; storage endianness and compression metadata are distinguished from the mathematical integers.

A Separating dependent radicals

A correlated-radical separation lemma

Let $F, D, Q, G, b, c \in \mathbb{Z}$, with $F \equiv D \equiv 1 \pmod 3$ and $Q \equiv 0 \pmod 3$, and put $\delta = Q - 1$. Suppose algebraic numbers satisfy

$$f^2 = F, \qquad \tau^2 = D, \qquad g^2 = G, \qquad \delta t^2 = Q(fb - c)^2 - 1.$$

If $\rho = f + 3\tau + 9t + 27g \in \mathbb{Q}$, then all four roots are rational. Thus f, τ, g are integers; if $Q = 0$ or $Q > 1$, then t is also an integer. The signs are unrestricted, and the dependence of t on f is intentional. No independence of square classes is assumed.

Proof. Let $R = \mathbb{Z}_{(3)}$ and let S be its integral closure in a finite normal extension of $\mathbb{Q}$ containing all four roots. The ring S is invariant under conjugation and $S \cap \mathbb{Q} = R$. Indeed, a reduced rational with denominator divisible by 3 cannot satisfy a monic equation with coefficients whose denominators are prime to 3: clearing denominators would contradict reduction modulo 3. The elements f, τ, g are integral over R. Since δ is a unit of R, the radicand of t belongs to S; transitivity of integrality then gives $t \in S$.

If f is irrational, an automorphism σ of the normal extension sends f to $-f$. Subtracting the conjugate rational relation gives

$$2f = 3(\sigma\tau - \tau) + 9(\sigma t - t) + 27(\sigma g - g).$$

Consequently $f/3 \in S$, since 2 is a unit in R. Squaring gives $F/9 \in S \cap \mathbb{Q} = R$, contradicting $F \equiv 1 \pmod 3$. Thus $f \in \mathbb{Q}$, hence $f \in \mathbb{Z}$. Apply the same argument to $(\rho - f)/3 = \tau + 3t + 9g$ to obtain $\tau \in \mathbb{Z}$. Now $t^2 = [Q(fb - c)^2 - 1]/\delta \in R$ is rational and congruent to 1 modulo 3. Applying the argument once more to $(\rho - f - 3\tau)/9 = t + 3g$ gives $t \in \mathbb{Q}$, and then $g \in \mathbb{Q}$, hence $g \in \mathbb{Z}$.

For the final integrality assertion, $Q = 0$ gives $t^2 = 1$. If $Q > 1$, set $u = fb - c \in \mathbb{Z}$ and write $|t| = a/m$ in lowest terms. The equation $Qu^2 - (Q-1)t^2 = 1$ gives $m^2 \mid Q - 1$. The cases $u = 0$ and $|u| = 1$ are respectively impossible and integral. Otherwise reflect signs so $u > 1, t > 0$; then $u < t < Qu/(Q-1)$. The norm-preserving transformation

$$u' = (2Q-1)u - 2(Q-1)t, \qquad t' = (2Q-1)t - 2Qu$$

has $u' \in \mathbb{Z}$, $0 < |u'| < u$, and leaves the reduced denominator m unchanged, since $2Q - 1 \equiv 1 \pmod m$. Descent to $|u| = 1$ proves $m = 1$. □

Scope. This is an ordinary algebraic lemma, including repeated or negative roots. It is not a new end-to-end consistency calibration. An elimination product must use the two t roots belonging to *each* selected sign of f; a four-independent-radical formula with a fixed t radicand would be wrong.

B The complete historical radix67 tuple

In this appendix K denotes the port K_0 in the new equations. Every symbol introduced here is a fixed constant, independent of the ten quantified variables. The formulas and 436-character seed are the retained indexed BFS943 radix67 specification; they reconstruct the exact supplied payloads, with $\beta = 2^{590069} - 1$.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ &\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ &\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\,\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934+3a+b-6\lfloor a/4 \rfloor$ $a \in \{0,1,4\},\ b=0,1,2$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\,\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

C The complete Start marker4177 tuple

Here K denotes the fixed port K_0, not an unknown. These are the whole coherent big-endian Start-marker4177 ports, not the historical radix67 tuple. **The four fixed integers.** Every index,

set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{4177}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{2059} + \rho^{3108}.$$

$$\begin{aligned} M_C &= \beta + \rho - (\rho+3)\rho^{990} \\ &\quad - G_{943}(\rho) - \Psi(\rho, 1), \\ M_F &= B + 3 + (2^{44} - 2)\Lambda \\ &\quad + \rho^{973} G_9(\rho) + \rho^{1012} G_8(\rho) + \rho^{1048} \\ &\quad + 2^{29}\rho^{4175} - \rho^{4157}, \\ K &= \rho^{20} + \rho^{30} + \rho^{1048} + \rho^{4187} \\ &\quad + \Lambda\Phi(1/\rho), \end{aligned}$$

where

$$\begin{aligned} \Phi(x) &= \sum_{i=0}^{942} [1 + 2^{10}(1 + w_i + 2^8 n_i \\ &\quad + 2^{17} p_i + 2^{25} s_i)] x^i + 2^{10}\Psi(x, 2), \\ \Psi(x, t) &= x^{992}(x^{10} + t^{17}) G_8(tx) \\ &\quad + t^8 x^{943}(x^{30} + t^{17}) G_9(tx) \\ &\quad + t^{34}(2 - t) x^{1048}. \end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2$ $(j = 2q + b < 430)$	$((1 - d_j) r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$434 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{424+2r_{d,q}+b \\ q \in \mathcal{S}_d,\ d,b=0,1}$	$(d r_{d,q}, b, (1 - d) r_{d,q}, h(q, b))$
$924 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$934 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b =$

$a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(E_0, \tau_q) - 14),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j)D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

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Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSD0Tgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```